Population Dynamics & Projections

Forecasting II: Bayesian Methods in Demography

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Readings

Books:

- Gelman, Carlin, Stern, Dunson, Vehtari and Rubin (2013). *Bayesian Data Analysis* (Third Edition). Chapman and Hall/CRC
- Gelman and Hill (2006). *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge University Press
- Hoff (2010). *A First Course in Bayesian Statistical Methods*. Springer

Articles:

- Fienberg (2006). "When Did Bayesian Inference Become 'Bayesian'?" *Bayesian Analysis*, 1(1), 1-40
- Wiśniowski, Smith, Bijak, Raymer and Forster (2015). "Bayesian Population Forecasting: Extending the Lee-Carter Method." *Demography*, 52(3), 1035–1059. Open Access. `https://doi.org/10.1007/s13524-015-0389-y`
- Ševčíková, H., Li, N., Kantorová, V., Gerland, P., Raftery, A.E. (2016). "Age-Specific Mortality and Fertility Rates for Probabilistic Population Projections." In: Schoen, R. (ed) *Dynamic Demographic Analysis*. The Springer Series on Demographic Methods and Population Analysis, vol 39. Springer, Cham. `https://doi.org/10.1007/978-3-319-26603-9_15`

Motivation

- Accurate measurements of demographic processes are essential for informing health and public policy decisions, as well as serving as the foundation for statistical inference across the social sciences.

The fundamental demographic balancing equation:

$$P(t+1) = P(t) + B(t, t+1) - D(t, t+1) + I(t, t+1) - O(t, t+1)$$

where B represents births, D deaths, I immigration, and O emigration during the period from t to $t+1$.

- With complete and accurate data on all these components, demography would be a straightforward accounting exercise.

Motivation

$$P(t+1) = P(t) + B(t, t+1) - D(t, t+1) + I(t, t+1) - O(t, t+1) + \text{ERROR}$$

- Need statistical methods to obtain reliable estimates
- The use of Bayesian methods has increased exponentially in the last decade
- Well-suited to demographic problems:
  - Often have prior knowledge about population patterns
  - Hierarchical structures
  - Need to reconcile data from different sources

Very Early Beginnings: Laplace

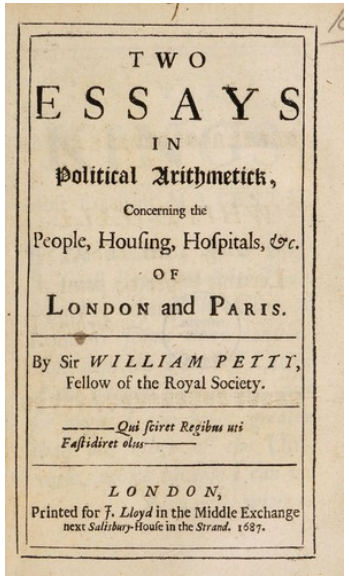
Figure: Pierre-Simon Laplace (1749-1827)

- Studying the sex ratio at birth in Paris (1781)
- Over 1745-1770, observed 251,527 boys and 241,945 girls
- Denote $x =$ probability that a given birth was male
- Laplace was interested in estimating $P(x \leq 0.5)$
- Using inverse probability he computed:

$$P(x \leq 0.5 \mid p = 251,527, q = 241,945) = 1.152 \times 10^{-42}$$

As it is exceedingly small, we can assert, with the same certainty as any other moral truth, that the difference observed in Paris between births of boys and those of girls is due to a greater likelihood for births of boys (Laplace 1781).

Very Limited Use of Statistical Methods in Demography for a Long Time



TWO
ESSAYS
IN
Political Arithmetick,
Concerning the
People, Housing, Hospitals, &c.
OF
LONDON and PARIS.

By Sir WILLIAM PETTY,
Fellow of the Royal Society.

——Qui sciret Regibus uti
Fastidiret otiis——

LONDON,
Printed for J. Lloyd in the Middle Exchange
next Salisbury-House in the Strand. 1687.

Figure: Frontispiece from Graunt's 'Natural and Political Observations' (1662)

- Influenced by 'political arithmetic' (Petty/Graunt)
- Emphasis on population-based registers and censuses
- Tabulation of population data for use in political, economic, and societal arguments
- Problems of variability and uncertainty largely ignored; errors documented but adjusted on a post-hoc basis

Bayesian Updating

Bayesian updating is a method of statistical inference that applies Bayes' theorem to revise prior beliefs when new data becomes available.

Bayes' theorem is given by:

$$
\begin{aligned}
p(\theta \mid y) &= \frac{p(y \mid \theta)\, p(\theta)}{p(y)} \\
&= \frac{p(y \mid \theta)\, p(\theta)}{\int p(y \mid \theta)\, p(\theta)\, d\theta} \\
&\propto p(y \mid \theta)\, p(\theta)
\end{aligned}
$$

Where:

- $p(\theta)$: prior belief about the parameter,
- $p(y \mid \theta)$: likelihood of observing the data,
- $p(\theta \mid y)$: updated belief (posterior),
- $p(y)$: marginal likelihood, ensuring the result is a valid probability distribution.

The Bayesian Approach (1)

- In the frequentist approach:
  - We start by collecting data.
  - These data are assumed to arise from a random process — whether from nature, measurement tools, survey designs, etc.
  - We specify a model for how the data are generated (e.g., independent and identically distributed, normally distributed, smooth density, linear trend, etc.).
  - This process is linked to a specific quantity of interest — such as a parameter or a distribution.
  - Although this quantity is unknown, it is treated as fixed. Our goal is to learn about it by:
    - Estimating its value,
    - Testing hypotheses about it,
    - Or quantifying uncertainty through tools like confidence intervals and p-values.

The Bayesian Approach (2)

- As in the frequentist framework, we begin by observing data, assumed to be generated by some random process.
- We make assumptions about this process (e.g., it follows a known parametric distribution), and link it to an unknown but fixed quantity of interest.
- However, instead of treating this quantity as fixed and unknown, we model it as a random variable — reflecting our uncertainty about its value.
- We begin with a prior distribution that represents our belief about this quantity before seeing the data.
- Once we observe the data, we use Bayes' theorem to update our belief — resulting in a posterior distribution that combines prior knowledge and evidence from the data.

The Bayesian Approach (3) — Example

- Suppose we are interested in estimating p, the proportion of women in a population.
- We randomly sample n individuals (with replacement), and record their gender: $X_1, \ldots, X_n$, where $X_i = 1$ if the person is a woman, and $X_i = 0$ otherwise.
- In the frequentist framework:
  - We estimate p using the sample mean (MLE),
  - Construct a confidence interval for p,
  - And possibly test a hypothesis (e.g., $H_0 : p = 0.496$ vs. $H_1 : p \neq 0.496$).
- In the Bayesian perspective:
  - We may believe, before seeing the data, that p is likely close to 0.5.
  - The Bayesian framework allows us to:
    1. Incorporate this prior belief mathematically,
    2. Update that belief based on the observed data.

The Bayesian Approach (4) — Example (Continued)

- We can express our prior belief about p quantitatively.
- For example:
  - We might believe there is a 90% chance that p lies between 0.4 and 0.6,
  - Or a 95% chance that it lies between 0.3 and 0.8.
- To model this belief, we treat p as if it were a random variable and assign it a probability distribution.
- Note: in reality, the true parameter p is fixed — it's our belief about its value that is uncertain.
- The Bayesian approach allows us to model this uncertainty using a prior distribution.
- For example, we can assume:

$$p \sim \text{Beta}(a, a) \quad \text{for some } a > 0$$

- This distribution is called the **prior distribution**.

The Bayesian Approach (5) — Example (Continued)

- In our statistical experiment, $X_1, \ldots, X_n$ are assumed to be i.i.d. Bernoulli random variables with parameter p, conditionally on p.
- After observing the sample $X_1, \ldots, X_n$, we update our belief about p by deriving its conditional distribution given the data.
- This updated distribution of p, given the data, is known as the **posterior distribution**.
- If the prior is $\text{Beta}(a, a)$, then the posterior distribution is:

$$\text{Beta}\left(a + \sum_{i=1}^{n} X_i,\ a + n - \sum_{i=1}^{n} X_i\right)$$

Bayes' Rule and the Posterior Distribution (1)

- Consider a parameter space Θ with a prior distribution given by a probability density function (pdf) $\pi(\cdot)$.
- Let $X_1, \ldots, X_n$ be a sample of n random variables.
- Denote by $p_n(\cdot \mid \theta)$ the joint pdf of $X_1, \ldots, X_n$ given θ, where $\theta \sim \pi$.
- It is typically assumed that $X_1, \ldots, X_n$ are i.i.d. conditional on θ.
- The conditional distribution of θ given the observed data $X_1, \ldots, X_n$ is called the **posterior distribution**.
- We denote this posterior density by $\pi(\cdot \mid X_1, \ldots, X_n)$.

Bayes' Rule and the Posterior Distribution (2)

- Bayes' formula states that:

$$\pi(\theta \mid X_1, \ldots, X_n) \propto \pi(\theta)\, p_n(X_1, \ldots, X_n \mid \theta), \quad \forall\, \theta \in \Theta$$

- The proportionality constant does not depend on θ. The full expression is:

$$\pi(\theta \mid X_1, \ldots, X_n) = \frac{\pi(\theta)\, p_n(X_1, \ldots, X_n \mid \theta)}{\int_\Theta p_n(X_1, \ldots, X_n \mid t)\, \mathrm{d}\pi(t)}, \quad \forall\, \theta \in \Theta$$

Bayes' Rule and the Posterior Distribution (3)

In the previous example:

- Prior (Beta distribution):

$$\pi(p) \propto p^{a-1}(1-p)^{a-1}, \quad p \in (0,1)$$

- Given p, the data $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Ber}(p)$, so the likelihood is:

$$p_n(X_1, \ldots, X_n \mid p) = p^{\sum_{i=1}^n X_i}(1-p)^{n-\sum_{i=1}^n X_i}$$

- Therefore, the unnormalized posterior is:

$$\pi(p \mid X_1, \ldots, X_n) \propto p^{a-1+\sum X_i}(1-p)^{a-1+n-\sum X_i}$$

- The posterior distribution is:

$$\text{Beta}\left(a + \sum_{i=1}^n X_i,\ a + n - \sum_{i=1}^n X_i\right)$$

Non-informative Priors (1)

- **Idea:** When prior knowledge is limited or absent, it is often desirable to use a prior that is as uninformative as possible.
- A common choice:

$$\pi(\theta) \propto 1$$

  That is, a constant probability density function over the parameter space Θ.
- If Θ is bounded, this corresponds to a uniform prior on Θ.
- If Θ is unbounded, the constant prior does **not** define a proper probability distribution (it is not normalizable).
- An **improper prior** is a nonnegative, measurable function $\pi(\cdot)$ over Θ that is not integrable.
- Despite being improper, such priors can still be used to derive valid posterior distributions using Bayes' theorem, provided the resulting posterior is proper.

Non-informative Priors (2) — Examples

Examples:

- If $p \sim \mathcal{U}(0,1)$ and, given p, the data $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Ber}(p)$, then:

$$\pi(p \mid X_1, \ldots, X_n) \propto p^{\sum X_i}(1-p)^{n-\sum X_i}$$

$$\text{i.e., the posterior is Beta}\left(1 + \sum_{i=1}^{n} X_i,\ 1 + n - \sum_{i=1}^{n} X_i\right)$$

- If $\pi(\theta) = 1$ for all $\theta \in \mathbb{R}$, and $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, 1)$, then:

$$\pi(\theta \mid X_1, \ldots, X_n) \propto \exp\left(-\frac{1}{2}\sum_{i=1}^{n}(X_i - \theta)^2\right)$$

$$\text{i.e., the posterior is } \mathcal{N}\left(\bar{X}_n,\ \frac{1}{n}\right)$$

Non-informative Priors (3) — Jeffreys Prior

- **Jeffreys prior:**

$$\pi_J(\theta) \propto \sqrt{\det I(\theta)}$$

  where $I(\theta)$ is the Fisher information matrix for the parameter θ in the model associated with $X_1, \ldots, X_n$ (when it exists).
- **In the previous examples:**
  - Example 1: $\pi_J(p) \propto \frac{1}{\sqrt{p(1-p)}}$, for $p \in (0,1) \rightarrow$ This corresponds to the prior Beta$(1/2, 1/2)$
  - Example 2: $\pi_J(\theta) \propto 1$, for $\theta \in \mathbb{R} \rightarrow$ This is an improper prior

Bayesian Estimation (2)

- **Bayes estimator:**

$$\hat{\theta}^{(\pi)} = \int_{\Theta} \theta \, \mathrm{d}\pi(\theta \mid X_1, \ldots, X_n)$$

  This quantity is known as the **posterior mean**.
- The Bayes estimator depends on the choice of the prior distribution π (hence the superscript π).

Bayesian Estimation (3)

- **In the previous examples:**
  - **Example 1:** With prior $\mathcal{B}(a, a)$, $a > 0$

$$\hat{p}^{(\pi)} = \frac{a + \sum_{i=1}^{n} X_i}{2a + n} = \frac{a/n + \bar{X}_n}{2a/n + 1}$$

  - In particular, with Jeffreys prior $a = \frac{1}{2}$

$$\hat{p}^{(\pi_J)} = \frac{\frac{1}{2n} + \bar{X}_n}{1/n + 1}$$

  - **Example 2:** With flat prior $\pi_J(\theta) \propto 1$

$$\hat{\theta}^{(\pi_J)} = \bar{X}_n$$

- In each case, the Bayes estimator is consistent and asymptotically normal.
- **In general:** The asymptotic properties of the Bayes estimator do not depend on the choice of the prior.

Population Projections

- World population projections produced by UN every two years since 1950
- Traditionally deterministic, many based on cohort-component models and matrix-based approaches ('book-keeping')

$$\begin{bmatrix} n_0 \\ n_1 \\ n_2 \\ \vdots \\ n_{\omega-1} \\ n_\omega \end{bmatrix}_{t+1} = \begin{bmatrix} F_0 & F_1 & F_2 & \cdots & F_{\omega-1} & F_\omega \\ S_0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & S_1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & S_{\omega-2} & 0 & 0 \\ 0 & 0 & \cdots & 0 & S_{\omega-1} & 0 \end{bmatrix} \begin{bmatrix} n_0 \\ n_1 \\ n_2 \\ \vdots \\ n_{\omega-1} \\ n_\omega \end{bmatrix}_t$$

Population projection matrix (Leslie matrix)

Population Projections

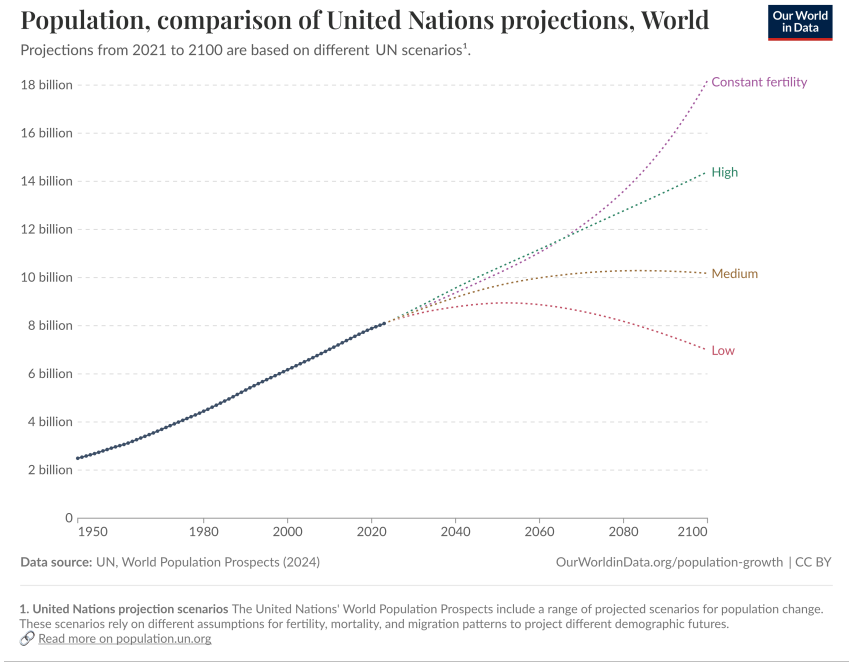


Figure: Comparison of United Nations population projections, World. Future projections onwards depict different UN fertility scenarios.

- UN high-medium-low variants to show possible ranges
- Constant fertility scenario for comparison
- Interest in ex-post 'how wrong were we?' (Keyfitz 1981) and 'shelf-life' of projections (Wilson 2019)

Some Foundational Discussions

- First widely-acknowledged stochastic forecasts made by the Finnish statistical office (Hyppola, 1949) using subjective opinions about fertility
- Seminal papers by Juha Alho (1985, 1990) on uncertainty in population forecasting
  A Bayesian analysis of the forecasting problem might be more elegant, in our opinion, because it would lead to a probability distribution for the future population given the past.
- Daponte et al (1997): Bayesian approach to combine expert opinion with classical projection methods
- Others from adjacent fields (e.g., Congdon) contributed to the development

The UN and Probabilistic Projections

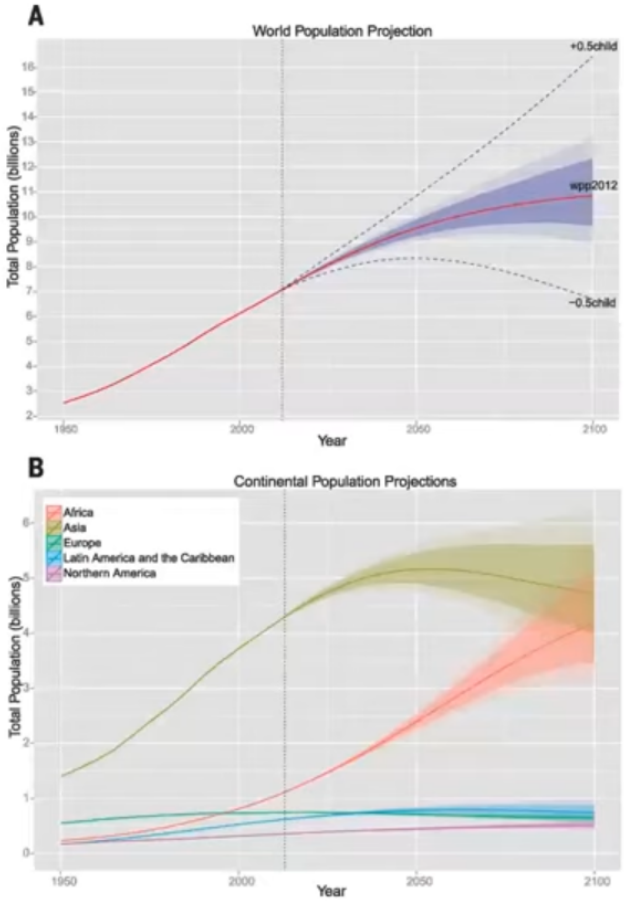


Figure: World Population Projection with probabilistic intervals

- In 2014, UN published probabilistic projections for the first time
- Led by Patrick Gerland at UNPD and Adrian Raftery at UW
- Based on Bayesian hierarchical models for fertility (Alkema et al 2011) and life expectancy (Raftery et al 2013)
- Hugely influential in changing how demographers think about uncertainty in projections

Many possibilities

- Paradigm shift in the field due to UN probabilistic projections
- Increasing computational power and ease of implementation
- Additionally:
  - Increased demand for timely global estimates due to MDGs/SDGs
  - Increasing demand for subnational estimates
  - Increased availability and use of non-traditional data

Applications - Bayesian Methods

1. **Projection**
   - Cohort component projection models
   - Fertility, mortality, and migration forecasting
2. **Incorporating expert opinion**
3. **Subnational (possibly small area) estimation**
4. **Combining and reconciling different data sources**
   - Social media data for demographic 'now-casting' (Alexander et al)

Applications - Bayesian Methods

- **Projection**
  - Cohort component projection models (Bryant, Wheldon, Alexander and Alkema)
- **Projection/forecasting of specific components of demographic change**
  - Fertility (Schmertmann et al)
  - Mortality (Girosi and King)
  - Migration (Raftery, Bijak, Wiśniowski et al)

Practical examples - R Lab (Friday)

This lab demonstrates how to generate probabilistic subnational projections for:

- Total Fertility Rate (TFR) using `bayesTFR`
- Life Expectancy at birth (e_0) using `bayesLife`
- Net Migration Rates using `bayesMig`

The methods follow:

- Ševčíková et al. (2018) for TFR
- Ševčíková and Raftery (2021) for life expectancy
- Azose and Raftery (2015, 2016) for migration

Data: County-level TFR, e_0, and migration rates (1990–2023); national projections from WPP2024.

Practical examples - R Lab (Today)

- Set up and starting (HMD and Rstan)
- Lee Carter Extension